Distributionally Robust LMI Control

Direct Data-Driven and DRC Synthesis

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Chapter 1

Distributionally Robust LMI Synthesis for LTI Systems

1.1 Problem Setting [1]

Output Feedback Control.

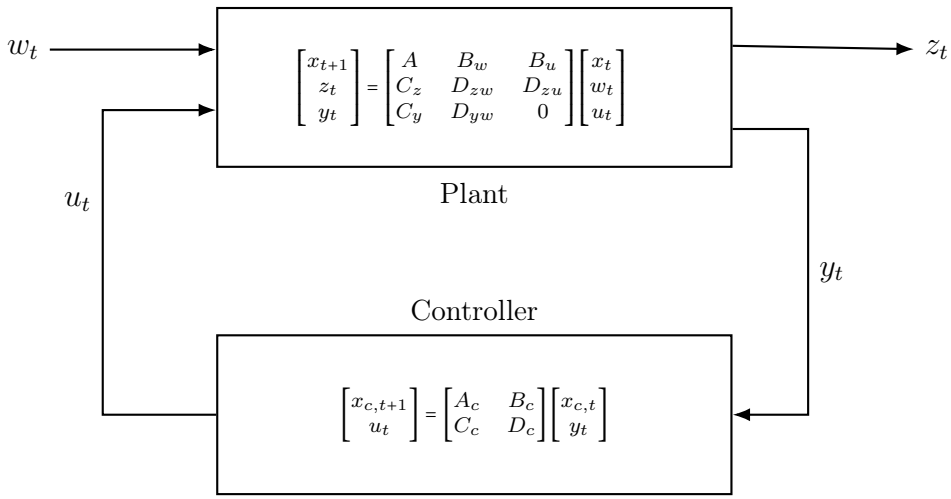


Figure 1.1: Closed-loop system with y feedback.

Define the combined state $\chi_t = [x_t^\top, x_{c,t}^\top]^\top$ and the closed-loop matrices

$$\mathcal{A} = \begin{bmatrix} A + B_u D_c C_y & B_u C_c \\ B_c C_y & A_c \end{bmatrix}, \quad \mathcal{B} = \begin{bmatrix} B_w + B_u D_c D_{yw} \\ B_c D_{yw} \end{bmatrix},$$
$$\mathcal{C} = \begin{bmatrix} C_z + D_{zu} D_c C_y & D_{zu} C_c \end{bmatrix}, \quad \mathcal{D} = D_{zw} + D_{zu} D_c D_{yw}.$$

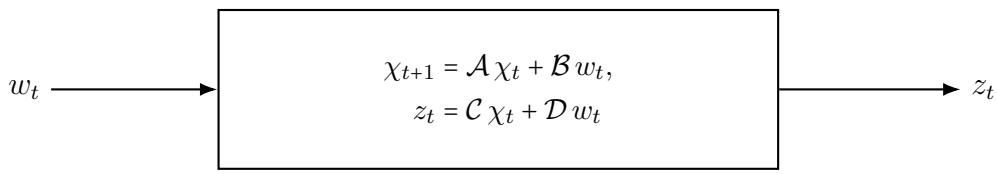


Figure 1.2: Lifted closed loop with state χ_t and matrices $(\mathcal{A}, \mathcal{B}, \mathcal{C}, \mathcal{D})$.

The disturbance formulation is formulated in Appendix A.

1.2 Paper-like Model Setting

The *paper-like* configuration [2] refers to a linear time-invariant (LTI) system of the form:

$$\begin{bmatrix} x_{t+1} \\ z_t \\ y_t \end{bmatrix} = \begin{bmatrix} A & B_w & B_u \\ C_z & D_{zw} & D_{zu} \\ C_y & D_{yw} & 0 \end{bmatrix} \begin{bmatrix} x_t \\ w_t \\ u_t \end{bmatrix},$$

where $x_t \in \mathbb{R}^{n_x}$ denotes the state, $u_t \in \mathbb{R}^{n_u}$ the control input, and $w_t \in \mathbb{R}^{n_w}$ an exogenous disturbance. The vectors $y_t \in \mathbb{R}^{n_y}$ and $z_t \in \mathbb{R}^{n_z}$ represent the measured and performance outputs, respectively.

The continuous-time version of this model describes a flexible turbine–shaft–pitch system:

$$\dot{x}(t) = A_{CT}x(t) + B_{CT}u(t) + E_{CT}w(t),$$

with

$$A_{CT} = \begin{bmatrix} \frac{k_\omega}{J} & \frac{k_h}{J} & 0 & \frac{k_\phi}{J} & 0 & \frac{k_\beta}{J} & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ \frac{f_\omega}{m} & -\omega_f^2 & -2\zeta_f\omega_f & 0 & 0 & \frac{f_\beta}{m} & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ \frac{m_\omega}{I_t} & 0 & 0 & -\omega_t^2 & -2\zeta_t\omega_t & \frac{m_\beta}{I_t} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 & -\omega_p^2 & -2\zeta_p\omega_p \end{bmatrix},$$

$$B_{CT} = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ \omega_p^2 \end{bmatrix}, \quad E_{CT} = \begin{bmatrix} \frac{k_v}{J} & 0 \\ 0 & 0 \\ \frac{f_v}{m} & \frac{f_v}{m} \\ 0 & 0 \\ \frac{m_v}{I_t} & \frac{m_v}{I_t} \\ 0 & 0 \\ 0 & 0 \end{bmatrix}.$$

which is discretized with sampling time T_s via

$$A = e^{A_{CT}T_s}, \qquad B_u = \int_0^{T_s} e^{A_{CT}\tau} B_{CT}\, d\tau, \qquad B_w = \int_0^{T_s} e^{A_{CT}\tau} E_{CT}\, d\tau.$$

The output matrices $(C_z, D_{zw}, D_{zu}, C_y, D_{yw})$ are selected to reflect a standard $\mathcal{H}_2$-type performance objective:

$$z_t = \begin{bmatrix} Q_x^{1/2} & 0 \\ 0 & R_u^{1/2} \end{bmatrix} \begin{bmatrix} x_t \\ u_t \end{bmatrix}, \qquad Q_x \succeq 0, R_u \succ 0 \qquad y_t = C_y x_t + D_{yw} w_t,$$

with $Q_x \succeq 0, R_u \succ 0$ and $D_{zw} = 0$.

1.3 Optimisation Problems [1]

Baseline

$$\min_{\theta \leftrightarrow (A_c, B_c, C_c, D_c)} J(\theta), \qquad \text{s.t. } \mathcal{A} \quad \text{isstable}$$

$$J(\theta) = \sup_{w \in \mathcal{W}} \quad \limsup_{T \to \infty} \frac{1}{T} \sum_{t=0}^{T-1} \mathbb{E} \, \|z_t\|^2 \tag{P1}$$

$$\text{s.t.} \quad \begin{aligned} \chi_{t+1} &= \mathcal{A} \chi_t + \mathcal{B} w_t, \\ z_t &= \mathcal{C} \chi_t + \mathcal{D} w_t \end{aligned}, \quad \chi_0 = 0$$

LMI

Introducing decision variables

$$\lambda \geq 0, \quad Q, \quad X = X^\top, \; Y = Y^\top, \; K, L, M, N,$$

and defining

$$\mathbb{P} = \begin{bmatrix} Y & I \\ I & X \end{bmatrix}, \qquad \mathbf{A} = \begin{bmatrix} AY + B_u M & A + B_u N C_y \\ K & XA + LC_y \end{bmatrix},$$

$$\mathbf{B} = \begin{bmatrix} B_w + B_u N D_{yw} \\ XB_w + LD_{yw} \end{bmatrix}, \quad \mathbf{C} = \begin{bmatrix} C_z Y + D_{zu} M & C_z + D_{zu} N C_y \end{bmatrix}, \quad \mathbf{D} = D_{zw} + D_{zu} N D_{yw}.$$

The objective bacame:

$$\min_{\lambda \geq 0, \; Q, \; X = X^\top, \; Y = Y^\top, \; K, L, M, N} \quad \operatorname{trace}(Q \Sigma_{\text{nom}}) + \lambda \gamma^2 \tag{P2}$$

subject to a block LMI that depends on the ambiguity model.

Correlated model ($\mathcal{W}_{\text{cor}}$). With identity blocks I_w, I_z of matching sizes, enforce

$$\begin{bmatrix} -\mathbb{P} & 0 & 0 & \mathbf{A}^\top & \mathbf{C}^\top \\ 0 & -\lambda I_w & \lambda I_w & \mathbf{B}^\top & \mathbf{D}^\top \\ 0 & \lambda I_w & -Q - \lambda I_w & 0 & 0 \\ \mathbf{A} & \mathbf{B} & 0 & -\mathbb{P} & 0 \\ \mathbf{C} & \mathbf{D} & 0 & 0 & -I_z \end{bmatrix} \prec 0, \quad \mathbb{P} \succeq 0.$$

Independent model ($\mathcal{W}_{\text{ind}}$). Impose the pair of LMIs

$$\begin{bmatrix} -\mathbb{P} & \mathbf{A}^\top & \mathbf{C}^\top \\ \mathbf{A} & -\mathbb{P} & 0 \\ \mathbf{C} & 0 & -I_z \end{bmatrix} \prec 0, \qquad \begin{bmatrix} -\lambda I_w & \lambda I_w & \mathbf{B}^\top & \mathbf{D}^\top \\ \lambda I_w & -Q - \lambda I_w & 0 & 0 \\ \mathbf{B} & 0 & -\mathbb{P} & 0 \\ \mathbf{D} & 0 & 0 & -I_z \end{bmatrix} \prec 0, \quad \mathbb{P} \succeq 0.$$

1.4 Direct Data-Driven DRO–LMI via Young-Type Relaxation

Idea. We first estimate the nominal plant maps from one persistently exciting trajectory $\{x_k, u_k, y_k, z_k\}_{k=0}^{T}$ by one–step regression. The remaining data mismatch is treated as an *uncertainty budget* and injected into the robust LMI through a Young-inequality relaxation. This keeps the synthesis convex while avoiding an explicit probabilistic model of the identification errors.

One–step estimates and residual decomposition

Form the stacked data matrices

$$X_- = \begin{bmatrix} x_0 & \cdots & x_{T-1} \end{bmatrix}, \qquad U_- = \begin{bmatrix} u_0 & \cdots & u_{T-1} \end{bmatrix}, \qquad X_+ = \begin{bmatrix} x_1 & \cdots & x_T \end{bmatrix}.$$
$$Y_- = \begin{bmatrix} y_0 & \cdots & y_{T-1} \end{bmatrix}, \qquad Z_- = \begin{bmatrix} z_0 & \cdots & z_{T-1} \end{bmatrix},$$

Dataset preprocessing. Rather than relying on a single open-loop experiment, which may lead to different simulation's results, multiple trajectories $\{\Gamma_i\}_{i=1}^{N=50}$ are collected, each containing matrices

$$\Gamma_i = \{X_i, U_i, Y_i, Z_i, X_{i,+}\}, \qquad X_i \in \mathbb{R}^{n_x \times T_i},\ U_i \in \mathbb{R}^{n_u \times T_i},\ Y_i \in \mathbb{R}^{n_y \times T_i},\ Z_i \in \mathbb{R}^{n_z \times T_i},$$

all truncated to a common horizon $T = \min_i T_i$. Each dataset is initialized with random states (i.e. $x_0 \sim \mathcal{N}(0, I)$) to ensure excitation diversity, and only trajectories satisfying the persistent–excitation condition $\sigma_{\min}\left(\frac{1}{T}\Phi\Phi^\top\right) > \varepsilon_{\mathrm{PE}}$ are kept before medoid selection.

Before selecting a nominal dataset, the trajectories are *normalized* to prevent one signal from dominating the others. For each variable $k \in \{X, U, Y, Z, X_+\}$, a scale factor s_k is computed as the mean Frobenius norm across seeds,

$$s_k = \frac{1}{N} \sum_{i=1}^{N} \|\Gamma_i[k]\|_F + \varepsilon,$$

and all data are rescaled as $\tilde{\Gamma}_i[k] = \Gamma_i[k]/s_k$. This ensures that signals with different physical units contribute comparably to the distance metric.

Among the N realizations, the most representative one (the *medoid*) is selected as

$$i^\star = \arg\min_i \sum_k w_k \sum_{j=1}^{N} \left\|\tilde{\Gamma}_j[k] - \tilde{\Gamma}_i[k]\right\|_F^2,$$

where w_k are optional weighting coefficients. The final matrices used for identification are then taken from this single dataset,

$$(X_-, U_-, Y_-, Z_-, X_+) = (X_{i^\star}, U_{i^\star}, Y_{i^\star}, Z_{i^\star}, X_{i^\star,+}).$$

Motivation. A naive arithmetic average of the trajectories, e.g. $\bar{X} = \frac{1}{N}\sum_i X_i$, is undesirable because opposite sign components can cancel, artificially shrinking the state amplitudes and destroying the dynamical consistency between X_- and X_+. The medoid procedure avoids this by selecting one real, physically consistent trajectory that is closest (in the Frobenius sense) to all others, effectively reducing the influence of measurement noise and random variations without corrupting the system structure.

SysID Define $D_x = \begin{bmatrix} X_- \\ U_- \end{bmatrix}$ and use the (regularized) right inverse $D_x^\dagger := D_x^\top (D_x D_x^\top + \rho I)^{-1}$ (with $\rho \geq 0$, $\rho = 10^{-6}$ in our implementation). The nominal state update is

$$\underbrace{\begin{bmatrix} \hat{A} & \hat{B}_u \end{bmatrix}}_{\hat{O}_x} = X_+ D_x^\dagger, \qquad \hat{A} \in \mathbb{R}^{n\times n},\ \hat{B}_u \in \mathbb{R}^{n\times m}.$$

Let the *state residuals* be

$$R := X_+ - \hat{A} X_- - \hat{B}_u U_- \in \mathbb{R}^{n\times T}.$$

We extract a low–rank disturbance channel from the principal directions of the residual covariance

$$S_R := \frac{1}{T} R R^\top, \qquad \tilde{S}_R := \tfrac{1}{2}\big(S_R + S_R^\top\big) + \varepsilon I_n = U_s \operatorname{diag}(s_1, \ldots, s_n)\, U_s^\top,$$

with eigenvalues ordered as $s_1 \geq \cdots \geq s_n \geq 0$.

Data–driven covariance modes. When the disturbance covariance is not known a priori, we choose the minimal rank n_w such that

$$n_w := \min\Big\{k : \sum_{i=1}^{k} s_i \ \geq\ \eta \sum_{i=1}^{n} s_i\Big\},$$

and define

$$\hat{B}_w\, \widehat{\Sigma}_w^{1/2} := U_s(:, 1{:}n_w) \operatorname{diag}\big(\sqrt{s_1}, \ldots, \sqrt{s_{n_w}}\big), \qquad \hat{W} := \hat{B}_w^\dagger R,$$

with $\widehat{\Sigma}_w := \operatorname{diag}(s_1, \ldots, s_{n_w})$ and nominal disturbance covariance

$$\Sigma_{\text{nom}} := \tfrac{1}{2}\Big(\tfrac{1}{T}\hat{W}\hat{W}^\top + \big(\tfrac{1}{T}\hat{W}\hat{W}^\top\big)^\top\Big) + \varepsilon I_{n_w}.$$

Known covariance mode. When a nominal disturbance covariance $\Sigma_{\text{nom}} \succ 0$ is known a priori, with dimension $n_w \times n_w$, we fix

$$n_w := \dim(\Sigma_{\text{nom}}).$$

Let

$$\Sigma_{\text{nom}} = Q \operatorname{diag}(\lambda_1, \ldots, \lambda_{n_w})\, Q^\top, \qquad \Sigma_{\text{nom}}^{-1/2} := Q \operatorname{diag}(\lambda_1^{-1/2}, \ldots, \lambda_{n_w}^{-1/2})\, Q^\top,$$

and denote by $U_p := U_s(:, 1:n_w)$ the eigenvectors of $\tilde{S}_R$ associated with the largest n_w eigenvalues $(s_1, \ldots, s_{n_w})$. We then set

$$\hat{B}_w := U_p \operatorname{diag}\big(\sqrt{s_1}, \ldots, \sqrt{s_{n_w}}\big)\, \Sigma_{\text{nom}}^{-1/2}.$$

By construction,

$$\hat{B}_w\, \Sigma_{\text{nom}}\, \hat{B}_w^\top = U_p \operatorname{diag}(s_1, \ldots, s_{n_w})\, U_p^\top,$$

so the dominant part of the residual covariance is matched using the prescribed disturbance covariance Σ_{nom}, and no additional re-estimation of Σ_{nom} from R is performed in this mode.

For the performance outputs we fix the weighting matrices as in the reference setup, defining $z = [\, Q_x^{1/2} x;\ R_u^{1/2} u \,]$ with

$$C_z = \begin{bmatrix} \mathrm{diag}(\sqrt{Q_x}) \\ 0_{n_u \times n_x} \end{bmatrix}, \quad D_{zu} = \begin{bmatrix} 0_{n_x \times n_u} \\ \mathrm{diag}(\sqrt{R_u}) \end{bmatrix}, \quad D_{zw} = 0, \quad Q_x = I_{n_x},\ R_u = I_{n_u}$$

so that the cost $\|z\|_2^2$ is consistent across the BL and DD formulations—this is a deliberate designer choice for fair comparison.

Algorithm 1 Dataset Preprocessing and Medoid Selection

Input: N trajectories $\Gamma_i = \{X_i, U_i, Y_i, Z_i, X_{i,+}\}$, time lengths T_i
Parameters: PE threshold $\varepsilon_{\mathrm{PE}}$, scaling offset ε, weights w_k
(A) Truncate and PE filtering
$T \leftarrow \min_i T_i$
for $i = 1$ to N **do**
    Truncate each Γ_i to horizon T
    Sample $x_0^{(i)} \sim \mathcal{N}(0, I)$
    Form $\Phi_i = [X_i; U_i]$
    if $\sigma_{\min}(\frac{1}{T}\Phi_i\Phi_i^\top) \le \varepsilon_{\mathrm{PE}}$ **then**
        discard Γ_i
(B) Normalization across datasets
for each block $k \in \{X, U, Y, Z, X_+\}$ **do**
    Compute scale $s_k = \frac{1}{N}\sum_{i=1}^{N} \|\Gamma_i[k]\|_F + \varepsilon$
    for $i = 1$ to N **do**
        $\tilde{\Gamma}_i[k] \leftarrow \Gamma_i[k]/s_k$
(C) Medoid selection
for $i = 1$ to N **do**
    Compute total normalized Frobenius distance:

$$d_i = \sum_k w_k \sum_{j=1}^{N} \|\tilde{\Gamma}_j[k] - \tilde{\Gamma}_i[k]\|_F^2$$

$i^\star = \arg\min_i d_i$
(D) Output
Representative dataset:

$$(X_-, U_-, Y_-, Z_-, X_+) = (X_{i^\star}, U_{i^\star}, Y_{i^\star}, Z_{i^\star}, X_{i^\star,+})$$

Algorithm 2 One–Step Linear Regression (SysID) and Residual Decomposition

Input: Representative dataset (X_-, U_-, X_+)

Parameters: regularization $\rho \geq 0$, energy fraction η, known/unknown covariance mode

(A) One-step regression

Form regression matrix $D_x = [X_-; U_-]$

Compute regularized right inverse:

$$D_x^\dagger = D_x^\top (D_x D_x^\top + \rho I)^{-1}$$

Estimate nominal dynamics:

$$[\, \hat{A} \; \hat{B}_u \,] = X_+ D_x^\dagger$$

(B) Residual computation

Compute state residuals:

$$R = X_+ - \hat{A} X_- - \hat{B}_u U_-$$

Residual covariance:

$$S_R = \tfrac{1}{T} R R^\top, \qquad \tilde{S}_R = \tfrac{1}{2}(S_R + S_R^\top) + \varepsilon I$$

Eigen-decomposition:

$$\tilde{S}_R = U_s \operatorname{diag}(s_1, \ldots, s_n)\, U_s^\top$$

with $s_1 \geq \cdots \geq s_n$.

(C) Disturbance-channel extraction

if unknown disturbance covariance **then**

Choose smallest n_w with

$$\sum_{i=1}^{n_w} s_i \geq \eta \sum_{i=1}^{n} s_i$$

Construct

$$\hat{B}_w \widehat{\Sigma}_w^{1/2} = U_s(:, 1{:}n_w) \operatorname{diag}(\sqrt{s_1}, \ldots, \sqrt{s_{n_w}})$$

Compute

$$\Sigma_{\text{nom}} = \tfrac{1}{2}\Big(\tfrac{1}{T} \hat{W} \hat{W}^\top + (\tfrac{1}{T} \hat{W} \hat{W}^\top)^\top \Big) + \varepsilon I$$

else

Use known $\Sigma_{\text{nom}} = Q \operatorname{diag}(\lambda_i) Q^\top$

$n_w = \dim(\Sigma_{\text{nom}})$

$U_p = U_s(:, 1{:}n_w)$

Build matched disturbance channel:

$$\hat{B}_w = U_p \operatorname{diag}(\sqrt{s_1}, \ldots, \sqrt{s_{n_w}})\, \Sigma_{\text{nom}}^{-1/2}$$

Output:

Nominal $(\hat{A}, \hat{B}_u)$, disturbance channel $\hat{B}_w$, residuals R, nominal covariance Σ_{nom}

A data–driven radius for model error

We bound the *identification error* on the state map with $\Delta_x \le RD^\dagger = \beta$ from $\hat{O}_x = X_+ D^\dagger := (O_x D)D^\dagger + RD^\dagger = O_x + \Delta_x$.

Here Δ_x collects the stacked deviations in $[A \;\; B_u]$ relative to $\hat{O}_x$. This bound is conservative but computable directly from data.

$$\begin{bmatrix} A & B_u \end{bmatrix} = \underbrace{\begin{bmatrix} \hat{A} & \hat{B}_u \end{bmatrix}}_{\hat{O}_x} + \Delta_x, \quad \Delta_x = \begin{bmatrix} \Delta_A & \Delta_B \end{bmatrix}, \quad \|\Delta_i\|_F \le \beta_i$$

with

$$\beta_A = \beta\sqrt{\frac{n_x}{n_x + n_u}}, \quad \beta_B = \beta\sqrt{\frac{n_u}{n_x + n_u}}$$

Therefore, writing the perturbed lifted block as $\mathbf{A}_{\text{true}} = \mathbf{A}(\hat{A}, \hat{B}_u) + E$, with $E \prec \beta_E$. Formal derivations and the precise kernel substitutions are given in Appendix B.1; the post–optimization reconstruction of the nominal maps from the recovered errors, $A \leftarrow \hat{A} + \Delta_A$, $B_u \leftarrow \hat{B}_u + \Delta_B$, is detailed in Appendix B.2.

Final program. The Young-relaxed data-driven synthesis (correlated case) is

$$\begin{aligned}
\min_{\substack{\lambda \ge 0,\ Q \ge 0,\\ S \ge 0\ \ P_s \ge 0\, H \ge 0\ \ G = \sigma_p I = G^T \ge 0\\ \sigma_s \ge 0,\ \sigma_p \ge 0,\ \{t\}_{i\in(K,L,M,N,X,Y)}\\ X = X^T,\ Y = Y^T,\ K,\ L,\ M,\ N}} \quad & \operatorname{tr}(Q\,\Sigma_{\text{nom}}) + \lambda\gamma^2 \ + \sum_{i\in(K,L,M,N,X,Y)} \rho\,(t_i + \varepsilon) \\
\text{s.t.}\ \ & P = P^T = \begin{bmatrix} Y & I \\ I & X \end{bmatrix} \succ 0, \quad \Xi(\mathbf{A},\mathbf{B},\mathbf{C},\mathbf{D}) \ \prec\ 0 \\
& \Xi(\mathbf{A},\mathbf{B},\mathbf{C},\mathbf{D})\Big|_{1,1} \leftarrow -P \ + \ (\sigma_s + \sigma_p)\ \beta_E \\
& \Xi(\mathbf{A},\mathbf{B},\mathbf{C},\mathbf{D})\Big|_{n,n} \leftarrow -P_s \\
& \begin{bmatrix} G & I \\ I & P \end{bmatrix} \succ 0, \qquad \begin{bmatrix} H & G \\ G & S \end{bmatrix} \succ 0, \qquad \begin{bmatrix} G + H & I \\ I & P_s \end{bmatrix} \succ 0 \\
& \sigma_s I - S \succ 0\ , \quad \begin{bmatrix} t_j I_{n^j_{row}} & j \\ j^\top & t_j I_{n^j_{col}} \end{bmatrix}\Bigg|_{j\in(K,L,M,N,X,Y)} \succeq 0.
\end{aligned}$$

where $\mathbf{A}, \mathbf{B}, \mathbf{C}, \mathbf{D}$ are built from the nominal data-based maps $(\hat{A}, \hat{B}, \hat{C}_y, \hat{D}_{yw}, \hat{C}_z, \hat{D}_{zu}, \hat{D}_{zw})$. For the correlated case, we have that $n = 4$, whereas in the independent one $n = 2$.

Appendix A

Disturbance Evaluation

The disturbance sequences w_t used in both the baseline and distributionally robust simulations are drawn from a *Wasserstein ambiguity set* [3, 4] centered at a nominal Gaussian model. The ambiguity set captures distributional uncertainty in the second-order statistics of the process noise, rather than assuming a fixed covariance.

Definition of the ambiguity set. For a given nominal covariance $\Sigma_{\text{nom}} \succ 0$ and Wasserstein radius $\gamma > 0$, the 2-Wasserstein ambiguity set is defined as

$$\mathcal{W}_2(\Sigma_{\text{nom}}, \gamma) = \Big\{ \mathcal{N}(0, \Sigma) \;\Big|\; W_2(\mathcal{N}(0, \Sigma), \mathcal{N}(0, \Sigma_{\text{nom}})) \le \gamma \Big\},$$

where $W_2(\cdot, \cdot)$ denotes the 2-Wasserstein distance between Gaussian measures:

$$W_2^2(\mathcal{N}(0, S_1), \mathcal{N}(0, S_2)) = \operatorname{tr}(S_1) + \operatorname{tr}(S_2) - 2 \operatorname{tr}\Big[\big(S_2^{1/2} S_1 S_2^{1/2} \big)^{1/2} \Big].$$

This set represents all Gaussian distributions whose covariance lies within distance γ (in the Bures metric) from the nominal covariance Σ_{nom}.

Sampling strategies. Two simulation modes are available:

1. **Independent (iid) disturbances:**

$$w_t \sim \mathcal{N}(0, \Sigma_{\text{nom}}), \quad t = 0, \dots, T-1,$$

   corresponding to temporally uncorrelated samples inside the ambiguity ball. Implementation uses a Cholesky factorization $L = \operatorname{chol}(\Sigma_{\text{nom}})$ and generates $w_t = L\, z_t,\ z_t \sim \mathcal{N}(0, I)$.

2. **Correlated disturbances:** To allow temporal correlation while preserving the same marginal covariance, an autoregressive model of order one (AR(1)) is simulated:

$$w_{t+1} = \rho\, w_t + \varepsilon_t, \qquad \varepsilon_t \sim \mathcal{N}(0, Q),$$

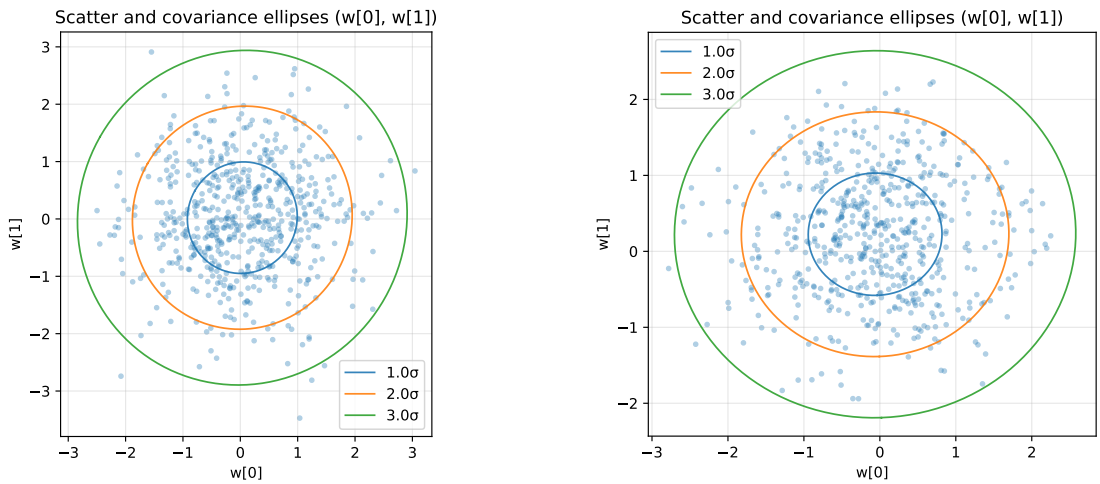
   with $0 < \rho < 1$ ensuring stability. The matrix Q is chosen so that

$$\Sigma_{\text{nom}} = \rho^2 \Sigma_{\text{nom}} + Q \quad \Rightarrow \quad Q = (1 - \rho^2)\Sigma_{\text{nom}}.$$

   This yields stationary correlated samples having marginal covariance Σ_{nom}, ensuring consistency with the ambiguity model.

In summary, the disturbances in simulation are not arbitrary noise sequences but samples from a *Wasserstein-consistent* uncertainty model. This maintains coherence between the data-driven identification, the LMI synthesis, and the subsequent Monte Carlo evaluation of closed-loop robustness.

Each figure below compares the *independent (iid)* and the *correlated (AR(1))* disturbance samplers that both lie in a 2-Wasserstein ball of radius γ around $\mathcal{N}(0, \Sigma_{\text{nom}})$. The iid and AR(1) samplers share the same stationary marginal $\mathcal{N}(0, \Sigma_{\text{nom}})$; they differ only in temporal structure (autocorrelation).



(a) Independent (iid): two-dimensional marginal samples (w_i, w_j) with nominal and empirical covariance ellipses. The empirical ellipse matches the nominal, confirming $\text{Cov}(w_t) = \Sigma_{\text{nom}}$ under iid sampling.

(b) Correlated (AR(1)): same marginal visualization. Marginals agree with Σ_{nom}; differences from iid are temporal, not marginal.

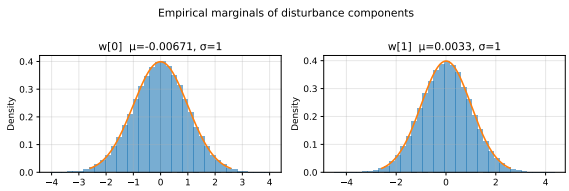
Figure A.1: Scatter and covariance ellipses. The panels visualize *marginal* covariance in the plane (w_i, w_j).

How iid vs correlated differ (and how they don't). Both modes satisfy $\mathbb{E}[w_t] = 0$ and $\text{Cov}(w_t) = \Sigma_{\text{nom}}$, hence the nearly identical scatter/ellipse and PDF panels. The difference is temporal:

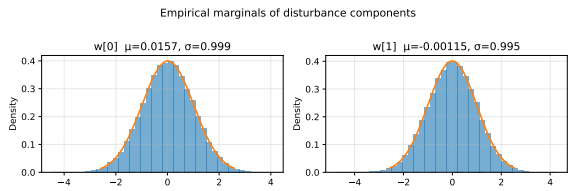
$$\text{iid:} \quad \text{Cov}(w_{t+k}, w_t) = 0 \ (k \neq 0), \qquad \text{AR(1):} \quad \text{Cov}(w_{t+k}, w_t) = \rho^{|k|}\Sigma_{\text{nom}}.$$

This is invisible in time-agnostic marginals. To expose it, add time-series diagnostics such as a trace of $w_i(t)$, an ACF plot (iid: spike at lag 0; AR(1): geometric decay ρ^k), or a lag-1 scatter (w_t, w_{t+1}) that aligns along the 45° line when correlated.

Technical caveat (boundary ellipses). The boundary covariances are constructed by eigenvalue shifts along principal axes of Σ_{nom}. This is exact in the commuting case (shared eigenvectors) and remains a useful visual surrogate otherwise; it illustrates the scale of admissible covariance perturbations at radius γ in the Bures/W_2 metric.

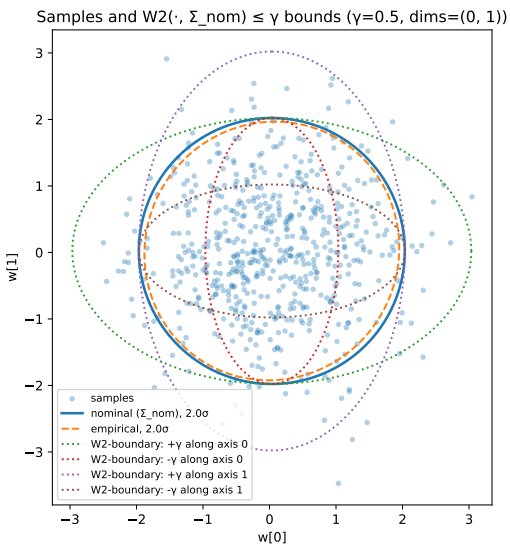


(a) Independent (iid): empirical marginal densities of disturbance components. Means are near zero and variances match the diagonal of Σ_{nom}.

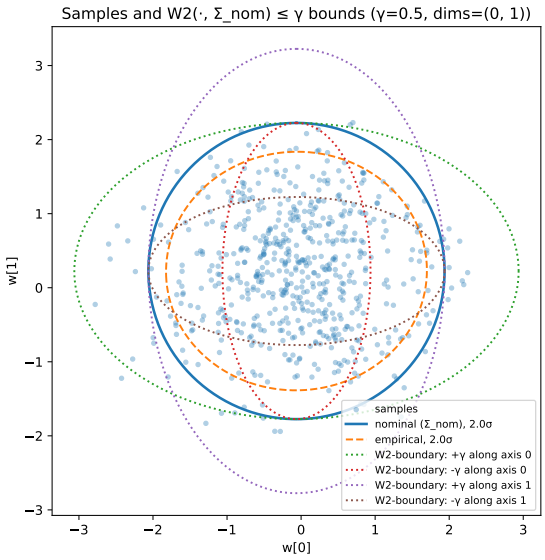


(b) Correlated (AR(1)): empirical marginal densities. As expected, they match the iid case; correlation does not alter stationary marginals.

Figure A.2: One-step marginal PDFs. Time order is ignored, so iid and AR(1) look nearly identical at the marginal level, as $T \to \inf$.



(a) Independent (iid): samples with nominal, empirical, and representative W_2-boundary ellipses. The nominal and empirical ellipses lie within the ball $W_2(\cdot, \Sigma_{\text{nom}}) \leq \gamma$.



(b) Correlated (AR(1)): correlated samples with the same W_2-boundary overlays. Correlation affects lag structure, not the W_2-marginal bounds.

Figure A.3: Wasserstein-ball overlays. Dotted ellipses illustrate covariances on the W_2 boundary around Σ_{nom}. They bound admissible *marginal* covariances, not individual points.

Gaussian ambiguity set and Wasserstein ball

We fix a nominal zero-mean Gaussian distribution

$$P_{\text{nom}} = \mathcal{N}(0, \Sigma_{\text{nom}}), \qquad \Sigma_{\text{nom}} \in \mathbb{S}_{++}^{n_w}.$$

For any zero-mean Gaussians $P_1 = \mathcal{N}(0, \Sigma_1)$, $P_2 = \mathcal{N}(0, \Sigma_2)$, the squared 2-Wasserstein (Gelbrich) distance is

$$W_2^2(P_1, P_2) = \operatorname{tr}(\Sigma_1 + \Sigma_2 - 2(\Sigma_2^{1/2}\Sigma_1\Sigma_2^{1/2})^{1/2}).$$

The closed Wasserstein ball of radius $\gamma > 0$ around P_{nom} is

$$\mathcal{B}_\gamma = \left\{\mathcal{N}(0, \Sigma) : \Sigma \in \mathbb{S}_{++}^{n_w},\ W_2\big(\mathcal{N}(0, \Sigma), \mathcal{N}(0, \Sigma_{\text{nom}})\big) \le \gamma\right\}.$$

All sampled disturbances are zero-mean Gaussians with covariance $\Sigma_{\text{use}} \in \mathcal{B}_\gamma$.

Projection onto the Wasserstein ball. Given a candidate covariance $\Sigma_{\text{raw}} \in \mathbb{S}_{++}^{n_w}$, we first check its distance to the nominal model:

$$d_{\text{raw}} = W_2(\mathcal{N}(0, \Sigma_{\text{raw}}), \mathcal{N}(0, \Sigma_{\text{nom}}))\,.$$

If $d_{\text{raw}} \le \gamma$, we keep $\Sigma_{\text{use}} = \Sigma_{\text{raw}}$. Otherwise, we move from Σ_{raw} towards Σ_{nom} along the Bures–Wasserstein geodesic.

Let

$$S = \Sigma_{\text{raw}}^{1/2}, \qquad A = \left(S\Sigma_{\text{nom}}S\right)^{1/2},$$

and define the geodesic

$$\Sigma(t) = \big((1-t)S + tA\big)^2, \qquad t \in [0, 1].$$

We then choose $t^\star \in (0, 1]$ such that

$$W_2(\mathcal{N}(0, \Sigma(t^\star)), \mathcal{N}(0, \Sigma_{\text{nom}})) = \gamma,$$

and set

$$\Sigma_{\text{use}} := \Sigma(t^\star).$$

In the code, $t^\star$ is found by a scalar bisection search on t.

Independent case ($\mathcal{W}_{\text{ind}}$). Given $\Sigma_{\text{use}} \in \mathcal{B}_\gamma$ and time horizon T, we generate an i.i.d. disturbance sequence

$$w_t \sim \mathcal{N}(0, \Sigma_{\text{use}}), \qquad t = 0, \dots, T-1,$$

independent across t. Equivalently, if L is a Cholesky factor $\Sigma_{\text{use}} = LL^\top$ and $z_t \sim \mathcal{N}(0, I_{n_w})$, then

$$w_t = Lz_t, \qquad t = 0, \dots, T-1.$$

This sequence belongs to the i.i.d. ambiguity set $\mathcal{W}_{\text{ind}}$.

Correlated case ($\mathcal{W}_{\mathbf{cor}}$). To obtain temporally correlated disturbances with stationary marginal $\mathcal{N}(0, \Sigma_{\text{use}})$, we use an AR(1) model

$$w_{t+1} = F w_t + \varepsilon_t, \qquad t = 0, \ldots, T-2,$$

with

$$F = \rho I_{n_w}, \qquad \varepsilon_t \sim \mathcal{N}(0, Q),$$

where $|\rho| < 1$ and Q is chosen so that the stationary covariance equals Σ_{use}. This is enforced by the discrete-time Lyapunov equation

$$\Sigma_{\text{use}} = F \Sigma_{\text{use}} F^\top + Q,$$

i.e.

$$Q = \Sigma_{\text{use}} - F \Sigma_{\text{use}} F^\top.$$

We initialize the process in stationarity,

$$w_0 \sim \mathcal{N}(0, \Sigma_{\text{use}}),$$

and then iterate the AR(1) recursion. The resulting trajectory $\{w_t\}_{t=0}^{T-1}$ has marginal distribution $\mathcal{N}(0, \Sigma_{\text{use}})$ at every t, hence lies in the correlated ambiguity set $\mathcal{W}_{\text{cor}}$.

Algorithm 3 Sampling Gaussian disturbances inside a 2-Wasserstein ball

Require: nominal covariance $\Sigma_{\text{nom}} \in \mathbb{S}_{++}^{n_w}$, radius $\gamma > 0$, time horizon T, mode `mode` $\in \{\text{independent}, \text{correlated}\}$, correlation coefficient $\rho \in (0, 1)$

Choose a candidate covariance $\Sigma_{\text{raw}} \in \mathbb{S}_{++}^{n_w}$, e.g. $\Sigma_{\text{raw}} = \begin{bmatrix} 0.69 & 0.53 \\ 0.53 & 1.61 \end{bmatrix}$
Compute $d_{\text{raw}} \leftarrow W_2(\mathcal{N}(0, \Sigma_{\text{raw}}), \mathcal{N}(0, \Sigma_{\text{nom}}))$
if $d_{\text{raw}} \le \gamma$ **then**
    $\Sigma_{\text{use}} \leftarrow \Sigma_{\text{raw}}$
else
    Compute $S \leftarrow \Sigma_{\text{raw}}^{1/2}$, $A \leftarrow (S \Sigma_{\text{nom}} S)^{1/2}$
    Define $\Sigma(t) \leftarrow ((1-t)S + tA)^2$ for $t \in [0, 1]$
    Find $t^\star \in (0, 1]$ by bisection such that $W_2(\mathcal{N}(0, \Sigma(t^\star)), \mathcal{N}(0, \Sigma_{\text{nom}})) = \gamma$
    $\Sigma_{\text{use}} \leftarrow \Sigma(t^\star)$
if `mode` = independent **then**
    Compute Cholesky factor $\Sigma_{\text{use}} = L L^\top$
    for $t = 0$ to $T-1$ **do**
        Draw $z_t \sim \mathcal{N}(0, I_{n_w})$
        $w_t \leftarrow L z_t$
else ▷ correlated AR(1) case
    $F \leftarrow \rho I_{n_w}$
    $Q \leftarrow \Sigma_{\text{use}} - F \Sigma_{\text{use}} F^\top$
    Compute Cholesky factors $\Sigma_{\text{use}} = L_\Sigma L_\Sigma^\top$, $Q = L_Q L_Q^\top$
    Draw $z_0 \sim \mathcal{N}(0, I_{n_w})$ and set $w_0 \leftarrow L_\Sigma z_0$
    for $t = 0$ to $T-2$ **do**
        Draw $\eta_t \sim \mathcal{N}(0, I_{n_w})$
        $\varepsilon_t \leftarrow L_Q \eta_t$
        $w_{t+1} \leftarrow F w_t + \varepsilon_t$
return $\{w_t\}_{t=0}^{T-1}$

Elliptical covariance visualization

For any zero-mean Gaussian distribution $\mathcal{N}(0, \Sigma) \subset \mathbb{R}^{n_w}$, its 2-dimensional marginal over coordinates (i, j) has covariance

$$\Sigma_{ij} := \begin{bmatrix} \Sigma_{ii} & \Sigma_{ij} \\ \Sigma_{ji} & \Sigma_{jj} \end{bmatrix} \in \mathbb{S}^2_{++}.$$

The *k-sigma covariance ellipse* associated with this marginal is the set

$$\mathcal{E}_k(\Sigma_{ij}) = \left\{ \mu_{ij} + V\Lambda^{1/2}Q : Q \in \mathbb{R}^2,\ \|Q\| \le k \right\},$$

where

$$\Sigma_{ij} = V\Lambda V^\top$$

is the eigenvalue decomposition, $\Lambda = \mathrm{diag}(\lambda_1, \lambda_2)$, $V = [v_1\ v_2]$ contains orthonormal eigenvectors, and k corresponds to the chosen confidence radius ($k = 2$ is a common visualization level).

Geometrically,

- v_1, v_2 give the orientation of the ellipse,
- $\sqrt{\lambda_1}, \sqrt{\lambda_2}$ give the axis lengths,
- μ_{ij} is the center (empirical sample mean).

Elements shown in the ellipse plots. Given sampled disturbances $w(t) \in \mathbb{R}^{n_w}$, the figure displays:

1. **Sample scatter:** Points $(w_i(t), w_j(t))$ for all t. These visualize the empirical marginal distribution.
2. **Nominal ellipse** (solid line): The covariance ellipse of the nominal model $\mathcal{N}(0, \Sigma_{\mathrm{nom}})$, projected to coordinates (i, j). This represents the center of the Wasserstein ball.
3. **Empirical ellipse** (dashed line): The ellipse of the empirical covariance of the samples. If the empirical marginal lies inside the Wasserstein ball, the ellipse must remain inside the corresponding boundary set.
4. **Wasserstein-boundary ellipses** (dotted lines): A set of synthetic covariances Σ_b lying on the boundary

   $$W_2\big(\mathcal{N}(0, \Sigma_b), \mathcal{N}(0, \Sigma_{\mathrm{nom}})\big) = \gamma.$$

   These are created by modifying eigenvalues of Σ_{nom} by $\pm\gamma$ along each principal axis. They visualize directions of maximal allowed covariance perturbation.

Axis-aligned boundary covariances. Let

$$\Sigma_{\text{nom}} = U\Lambda U^\top \quad \text{with} \quad \Lambda = \text{diag}(\lambda_1, \ldots, \lambda_{n_w}),\ \lambda_k > 0.$$

Define the nominal square roots

$$s_k = \sqrt{\lambda_k}.$$

To generate illustrative points on the Wasserstein boundary, we modify a single axis at a time:

$$s'_k = \max(0,\ s_k \pm \gamma), \qquad \lambda'_k = (s'_k)^2,$$

and keep all other eigenvalues unchanged.

Each boundary covariance is then

$$\Sigma_b = U \begin{bmatrix} \lambda'_1 & & \\ & \ddots & \\ & & \lambda'_{n_w} \end{bmatrix} U^\top,$$

and satisfies $W_2(\mathcal{N}(0, \Sigma_b), \mathcal{N}(0, \Sigma_{\text{nom}})) = \gamma$ in the commutative-eigenvector case.

Algorithm 4 Plotting marginal ellipses and Wasserstein bounds

---

Require: samples $w \in \mathbb{R}^{T \times n_w}$, nominal covariance Σ_{nom}, Wasserstein radius γ, coordinate pair (i, j), confidence level k (e.g. $k = 2$)

1: Extract 2D samples $W = w[:, (i, j)]$
2: Compute empirical mean μ and sample covariance Σ_{emp}
3: Compute 2D nominal covariance $\Sigma_{\text{nom},2d} = \Sigma_{\text{nom}}[(i, j), (i, j)]$
4: Generate a small set of boundary covariances:

$$\{\Sigma_b\} \leftarrow \text{axis-wise eigenvalue shifts of } \Sigma_{\text{nom}} \text{ by } \pm\gamma$$

5: For each covariance $\Sigma \in \{\Sigma_{\text{nom},2d}, \Sigma_{\text{emp},2d}, \Sigma_b^{(i,j)}\}$:

1. Compute eigen-decomposition $\Sigma = V\Lambda V^\top$
2. Generate ellipse points

$$\mathcal{E}_k = \left\{ \mu + V\Lambda^{1/2} \begin{bmatrix} \cos\theta \\ \sin\theta \end{bmatrix} k : \theta \in [0, 2\pi] \right\}$$

3. Add to the plot using:
   - solid line for nominal Σ_{nom}
   - dashed line for empirical Σ_{emp}
   - dotted lines for boundary ellipses Σ_b

6: Scatter-plot the sample points W
7: Return rendered figure

Appendix B

Dimonstrantion

B.1 Closed-loop lifting and decision variables

We keep the standard output-feedback change of variables

$$P = \begin{bmatrix} Y & I \\ I & X \end{bmatrix} \succ 0, \qquad \begin{aligned} \mathbf{A} &= \begin{bmatrix} \hat{A}Y + \hat{B}_u M & \hat{A} + \hat{B}_u N \hat{C}_y \\ K & X\hat{A} + L\hat{C}_y \end{bmatrix}, \\ \mathbf{B} &= \begin{bmatrix} \hat{B}_w + \hat{B}_u N \hat{D}_{yw} \\ X\hat{B}_w + L\hat{D}_{yw} \end{bmatrix}, \quad \mathbf{C} = \begin{bmatrix} \hat{C}_z Y + \hat{D}_{zu} M & \hat{C}_z + \hat{D}_{zu} N \hat{C}_y \end{bmatrix}, \\ \mathbf{D} &= \hat{D}_{zw} + \hat{D}_{zu} N \hat{D}_{yw}. \end{aligned}$$

The decision set is $X = X^\top,\ Y = Y^\top,\ K,\ L,\ M,\ N,\ Q,\ \lambda \geq 0$ and the Wasserstein-DRO objective is

$$\min \quad \operatorname{tr}(Q\,\Sigma_{\text{nom}}) + \lambda\gamma^2.$$

Setup. Start from the correlated-noise kernel

$$\Xi(\mathbf{A}, \mathbf{B}, \mathbf{C}, \mathbf{D}) := \begin{bmatrix} -P & 0 & 0 & \mathbf{A}^\top & \mathbf{C}^\top \\ 0 & -\lambda I & \lambda I & \mathbf{B}^\top & \mathbf{D}^\top \\ 0 & \lambda I & -Q - \lambda I & 0 & 0 \\ \mathbf{A} & \mathbf{B} & 0 & -P & 0 \\ \mathbf{C} & \mathbf{D} & 0 & 0 & -I \end{bmatrix} \prec 0, \tag{$\star$}$$

with decision variables $P \succ 0$, $Q \succeq 0$, $\lambda \geq 0$, and where $\mathbf{A}, \mathbf{B}, \mathbf{C}, \mathbf{D}$ are the usual lifted blocks built from the controller variables (X, Y, K, L, M, N) and the plant maps.

1. **Isolate the Lyapunov generator.** Focus on the $(1,4)$–$(4,1)$ corner that produces the Lyapunov term:

$$M(\mathbf{A}_{\text{true}}, P) := \begin{bmatrix} -P & \mathbf{A}_{\text{true}}^\top \\ \mathbf{A}_{\text{true}} & -P \end{bmatrix} \prec 0.$$

Starting from the Schur complement of the $(4,4)$ block $-P$ we obtain

$$-P - \mathbf{A}_{\text{true}}^\top (-P)^{-1} \mathbf{A}_{\text{true}} = -P + \mathbf{A}_{\text{true}}^\top P^{-1} \mathbf{A}_{\text{true}} \prec 0. \tag{1}$$

This form expresses discrete-time stability in the inverse-P coordinates.

2. **Inject identification error.** Let $A = \hat{A} + \Delta_A$, $B_u = \hat{B}_u + \Delta_B$, with data-driven bounds $\|\Delta_A\|_F \le \beta_A$, $\|\Delta_B\|_F \le \beta_B$. In the lifted block, $\mathbf{A}$ depends *affinely* on (A, B_u), so replacing by $\hat{A}, \hat{B}_u$ plus errors gives

$$\mathbf{A}_{\text{true}} = \mathbf{A}(\hat{A}, \hat{B}_u) + E_A + E_B,$$

where E_A collects the contributions linear in Δ_A, and E_B those linear in Δ_B. In particular:

$$E_A = \begin{bmatrix} \Delta_A Y & \Delta_A \\ 0 & X\Delta_A \end{bmatrix} \qquad E_B = \begin{bmatrix} \Delta_B M & \Delta_B N C_y \\ 0 & 0 \end{bmatrix} \tag{B.1}$$

Expanding (L) with $\mathbf{A}_{\text{true}}$, with $E = E_A + E_B$, gives

$$\mathbf{A}^\top P \mathbf{A} - P \; + \; \underbrace{\mathbf{A}^\top P E + E^\top P \mathbf{A}}_{\text{mixed terms}} \; + \; \underbrace{E^\top P E}_{\text{2nd order}} \prec 0,$$

where $\mathbf{A} := \mathbf{A}(\hat{A}, \hat{B}_u)$ is the nominal lifted block.

3. **Bound on the perturbation block E.** We start from the base bound

$$\beta = RD^\dagger \in \mathbb{R}^{(n_x+n_u)\times(n_x+n_u)},$$

and split it between the state and input parts as

$$\beta_A = \beta\sqrt{\frac{n_x}{n_x+n_u}}, \qquad \beta_B = \beta\sqrt{\frac{n_u}{n_x+n_u}}.$$

The corresponding spectral–norm bounds are

$$\kappa_A := \|\beta_A\|_2, \qquad \kappa_B := \|\beta_B\|_2,$$

and we assume

$$\|\Delta_A\|_2 \le \kappa_A, \qquad \|\Delta_B\|_2 \le \kappa_B.$$

The lifted perturbation block is

$$E \;=\; \begin{bmatrix} \Delta_A Y + \Delta_B M & \Delta_A + \Delta_B N C_y \\ 0 & X\Delta_A \end{bmatrix},$$

where the design matrices satisfy the norm bounds

$$\|Y\|_2 \le t_Y, \qquad \|M\|_2 \le t_M, \qquad \|N\|_2 \le t_N, \qquad \|X\|_2 \le t_X,$$

and C_y is constant with $c_y := \|C_y\|_2$.

Using submultiplicativity and the triangle inequality, we obtain

$$\|\Delta_A Y + \Delta_B M\|_2 \le \|\Delta_A\|_2\|Y\|_2 + \|\Delta_B\|_2\|M\|_2 \le \kappa_A t_Y + \kappa_B t_M =: e_{11},$$

$$\|\Delta_A + \Delta_B N C_y\|_2 \le \|\Delta_A\|_2 + \|\Delta_B\|_2\|N\|_2\|C_y\|_2 \le \kappa_A + \kappa_B t_N c_y =: e_{12},$$

$$\|X\Delta_A\|_2 \le \|X\|_2\|\Delta_A\|_2 \le t_X \kappa_A =: e_{22}.$$

A conservative global bound on E follows from

$$\|E\|_2^2 \;\le\; \|E\|_F^2 \;\le\; e_{11}^2 + e_{12}^2 + e_{22}^2 =: \gamma_E,$$

which implies

$$E^\top E \;\le\; \beta_E \quad \text{with} \quad \beta_E := \gamma_E I = \left(e_{11}^2 + e_{12}^2 + e_{22}^2\right) I.$$

This constant matrix β_E is then used as an upper bound for the perturbation energy in the subsequent LMIs.

4. Robustification of the Lyapunov inequality under model error.

Starting from

$$M(\mathbf{A}_{\text{true}}, P) = \begin{bmatrix} -P & \mathbf{A}_{\text{true}}^\top \\ \mathbf{A}_{\text{true}} & -P \end{bmatrix} \prec 0 \quad \Longleftrightarrow \quad -P + \mathbf{A}_{\text{true}}^\top P^{-1} \mathbf{A}_{\text{true}} \prec 0,$$

insert the decomposition

$$\mathbf{A}_{\text{true}} = \mathbf{A} + E, \qquad E = E_A + E_B.$$

Expanding gives

$$-P + \mathbf{A}^\top P^{-1} \mathbf{A} + E^\top P^{-1} E + E^\top P^{-1} \mathbf{A} + \mathbf{A}^\top P^{-1} E \prec 0. \tag{3.1}$$

Bounding the inverse metric. Introduce $\sigma_p > 0$ such that

$$P^{-1} \prec \sigma_p I. \tag{3.2}$$

This is equivalent to

$$\begin{bmatrix} \sigma_p I & I \\ I & P \end{bmatrix} \succ 0. \tag{3.3}$$

Thus

$$E^\top P^{-1} E \preceq \sigma_p E^\top E, \qquad E^\top E \preceq B_E \quad \Rightarrow \quad E^\top P^{-1} E \preceq \sigma_p B_E. \tag{3.5}$$

Bounding the mixed term. For any $S \succ 0$,

$$E^\top P^{-1} \mathbf{A} + \mathbf{A}^\top P^{-1} E = 2\langle E,\; P^{-1}\mathbf{A}\rangle \;\le\; E^\top S E + \mathbf{A}^\top P^{-1} S^{-1} P^{-1} \mathbf{A}. \tag{3.6}$$

Introduce $\sigma_s > 0$ with

$$S \prec \sigma_s I \quad \Longleftrightarrow \quad \sigma_s I - S \succ 0. \tag{3.7}$$

Then

$$E^\top S E \preceq \sigma_s B_E. \tag{3.8}$$

Combine all bounds. Using (3.5), (3.6), (3.8):

$$-P + \mathbf{A}^\top\big(P^{-1} - P^{-1}SP^{-1}\big)\mathbf{A} + \sigma_p B_E + \sigma_s B_E \prec 0. \tag{3.9}$$

Define

$$\bar{P} := P - \sigma_e B_E, \quad \sigma_e = \sigma_s + \sigma_p, \tag{3.10}$$

so $P_s \succ 0$. Its inverse satisfies

$$P_s^{-1} = P^{-1} + P^{-1}SP^{-1} \succ 0 \tag{3.11}$$

Therefore

$$-\bar{P} + \mathbf{A}^\top P_s^{-1}\mathbf{A} \prec 0 \quad \Longleftrightarrow \quad \begin{bmatrix} -\bar{P} & \mathbf{A}^\top \\ \mathbf{A} & -P_s \end{bmatrix} \prec 0. \tag{3.12}$$

To bound P_s^{-1} we do the following.

Auxiliary lifting. Define

$$G = \sigma_p I = G^\top \succ 0, \qquad G \succ P^{-1}, \tag{3.13}$$

and introduce slack $H \succ 0$ via

$$\begin{bmatrix} H & G \\ G & S \end{bmatrix} \succ 0 \quad \Longleftrightarrow \quad H - GS^{-1}G \succ 0. \tag{3.15}$$

Final condition. The robust requirement becomes

$$G + H - P_s^{-1} \succ 0, \tag{3.16}$$

equivalently,

$$\begin{bmatrix} G - H & I \\ I & P_s \end{bmatrix} \succ 0. \tag{3.17}$$

And thus, $P_s^{-1} = P^{-1} + P^{-1}S^{-1}P^{-1} \prec G + GS^{-1}G \prec G + H$.

5. **Regularization term (convex surrogate).**

To control the operator norms of the learned matrices K, L, M, N, introduce auxiliary scalar variables $t_K, t_L, t_M, t_N \geq 0$ and impose the epigraph LMIs

$$\begin{bmatrix} t_K I_{n_x} & K \\ K^\top & t_K I_{n_x} \end{bmatrix} \succeq 0, \qquad \begin{bmatrix} t_L I_{n_x} & L \\ L^\top & t_L I_{n_y} \end{bmatrix} \succeq 0, \tag{B.2}$$

$$\begin{bmatrix} t_M I_{n_u} & M \\ M^\top & t_M I_{n_x} \end{bmatrix} \succeq 0, \qquad \begin{bmatrix} t_X I_{n_x} & X \\ X^\top & t_X I_{n_x} \end{bmatrix} \succeq 0, \tag{B.3}$$

$$\begin{bmatrix} t_Y I_{n_x} & Y \\ Y^\top & t_Y I_{n_x} \end{bmatrix} \succeq 0, \qquad \begin{bmatrix} t_N I_{n_u} & N \\ N^\top & t_N I_{n_y} \end{bmatrix} \succeq 0. \tag{B.4}$$

These LMIs ensure $\|K\|_2 \leq t_K$, $\|L\|_2 \leq t_L$, $\|X\|_2 \leq t_X$, $\|Y\|_2 \leq t_y$, $\|M\|_2 \leq t_M$, and $\|N\|_2 \leq t_N$.

A convex spectral-norm regularization term is then added to the objective:

$$\mathcal{R}_{\text{spec}} = \rho\,(t_K + t_L + t_M + t_N + t_X + t_Y),$$

where the weights $\rho > 0$ tune the relative penalization of each block.

Optionally, a small numerical offset $\varepsilon > 0$ can be included to improve conditioning, yielding

$$\mathcal{R}_{\text{spec},\varepsilon} = \rho\,[(t_K + \varepsilon) + (t_L + \varepsilon) + (t_M + \varepsilon) + (t_M + \varepsilon) + (t_X + \varepsilon) + (t_Y + \varepsilon)].$$

Together with the previous convex surrogate term $\mathcal{R}_\varepsilon$, these define the complete regularized optimization problem, with $\mathcal{R} = \mathcal{R}_\varepsilon + \mathcal{R}_{\text{spec},\varepsilon}$:

$$\min_{\text{decision vars}} \; J_{\text{main}} + \mathcal{R} \quad \text{s.t.} \quad \text{LMI constraints above.}$$

B.2 Post-optimization recovery of Δ_A and Δ_B

Inputs. After solving the program, suppose we have optimal quantities

$$P^\star \succ 0, \quad X^\star = X^{\star\top}, \quad Y^\star = Y^{\star\top}, \quad M^\star,\ N^\star,\ s_A^\star > 0,\ s_B^\star > 0,\ \tau_A^\star,\ \tau_B^\star,$$

together with nominal identified maps $\hat{A}$, $\hat{B}_u$, $\hat{C}_y$ and data-driven radii β_A, β_B that induce the lifted bounds

$$\|E_A\|_F \le \beta_A := \sqrt{1 + \|X^\star\|_2^2 + \|Y^\star\|_2^2}\,\beta_A, \qquad \|E_B\|_F \le \beta_B := \sqrt{\|M^\star\|_2^2 + \|N^\star\|_2^2\|\hat{C}_y\|_2^2}\,\beta_B.$$

Step 1: Adversarial directions for the Young bounds. Let the nominal lifted block (built with $\hat{A}, \hat{B}_u$) be

$$\mathbf{A}^\star := \mathbf{A}(\hat{A}, \hat{B}_u; X^\star, Y^\star, M^\star, N^\star), \qquad Z_{A,i}^\star := P^\star \mathbf{A}^\star, \quad i \in \{A, B\}.$$

Choose worst-case aligned perturbations that saturate the Young relaxation:

$$E_{A,i}^\star = \begin{cases} \beta_{A,i}\dfrac{Z_{A,i}^\star}{\|Z_{A,i}^\star\|_F}, & Z_{A,i}^\star \neq 0, \\ 0, & Z_{A,i}^\star = 0, \end{cases} \qquad \beta_{A,A} := \beta_A, \quad \beta_{A,B} := \beta_B. \tag{B.5}$$

This choice attains equality in Young's inequality when $s_{A,i}^\star = \|Z_{A,i}^\star\|_F/\beta_{A,i}$.

Step 2: Block structure of $E_A^\star$ and $E_B^\star$. Recall the definitions

$$E_A = \begin{bmatrix} \Delta_A Y^\star & \Delta_A \\ 0 & X^\star \Delta_A \end{bmatrix}, \qquad E_B = \begin{bmatrix} \Delta_B M^\star & \Delta_B N^\star \hat{C}_y \\ 0 & 0 \end{bmatrix}.$$

After choosing $E_A^\star, E_B^\star$ via (B.5), extract the nonzero sub-blocks

$$A_1 := E_A^\star(1{:}n,\ 1{:}n), \quad A_2 := E_A^\star(1{:}n,\ n{+}1{:}2n), \quad A_3 := E_A^\star(n{+}1{:}2n,\ n{+}1{:}2n),$$

$$B_1 := E_B^\star(1{:}n,\ 1{:}m), \qquad B_2 := E_B^\star(1{:}n,\ m{+}1{:}m{+}p),$$

Step 3: Recovery of Δ_A (Sylvester equation). Solve the least-squares normal equation

$$\boxed{(X^{\star\top}X^{\star})\,\Delta_A \;+\; \Delta_A\,(Y^{\star}Y^{\star\top}+I) \;=\; A_2 + A_1Y^{\star\top} + X^{\star\top}A_3\ ,} \tag{B.6}$$

which is a Sylvester equation in Δ_A. Under mild conditions on the spectra of $X^{\star\top}X^{\star}$ and $Y^{\star}Y^{\star\top}+I$, (B.6) has a unique solution.

Special cases. If $X^{\star}=0$, then $\Delta_A(Y^{\star}Y^{\star\top}+I) = A_2+A_1Y^{\star\top} \Rightarrow \Delta_A = (A_2+A_1Y^{\star\top})(Y^{\star}Y^{\star\top}+I)^{-1}$. If $Y^{\star}=0$, then $(X^{\star\top}X^{\star}+I)\Delta_A = A_2 + X^{\star\top}A_3 \Rightarrow \Delta_A = (X^{\star\top}X^{\star}+I)^{-1}(A_2+X^{\star\top}A_3)$.

Step 4: Recovery of Δ_B (right-side Gram solve). Solve

$$\boxed{\Delta_B\,G_B \;=\; B_1M^{\star\top} + B_2\hat{C}_y^{\top}N^{\star\top}, \qquad G_B := M^{\star}M^{\star\top} + N^{\star}\hat{C}_y\hat{C}_y^{\top}N^{\star\top}\ ,} \tag{B.7}$$

so that, if G_B is invertible, $\Delta_B = (B_1M^{\star\top} + B_2\hat{C}_y^{\top}N^{\star\top})\,G_B^{-1}$. If G_B is singular or ill-conditioned, use the Moore–Penrose pseudoinverse G_B^{+} or a Tikhonov-regularized inverse $(G_B+\eta I)^{-1}$ with small $\eta>0$.

Step 5: Augmented nominal maps. Finally set

$$A_{\text{aug}} = \hat{A} + \Delta_A, \qquad B_{u,\text{aug}} = \hat{B}_u + \Delta_B,$$

and, if desired, verify feasibility with the identification radii: $\|\Delta_A\|_F \le \beta_A$, $\|\Delta_B\|_F \le \beta_B$ and with the lifted bounds $\|E_A(\Delta_A)\|_F \le \beta_A$, $\|E_B(\Delta_B)\|_F \le \beta_B$.

Remarks. (i) The recovered (Δ_A, Δ_B) are *representative* worst-case directions induced by the Young relaxation, not the unique identification errors. (ii) Numerically, (B.6) is solved with a Sylvester solver; (B.7) is a right-multiplication by a Gram factor and should use a robust linear solver or SVD-based pseudoinverse. (iii) If a single uncertainty block on $[A\ B_u]$ was used in synthesis, replace β_A, β_B by a common β and set $E_x^{\star} = \beta\, Z_x^{\star}/\|Z_x^{\star}\|_F$ with $Z_x^{\star} = P^{\star}[\hat{A}\ \hat{B}_u]$, then project $E_x^{\star}$ onto the E_A, E_B structures above before solving (B.6)–(B.7).

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